

Show all work. Indicate if values determined via shock table or code, if not otherwise obvious from the work. Submit any code used. If an LLM is used, attach a separate chat thread for each problem.

ME 561 students are required to typeset assignments using $\text{\LaTeX}$. **ME 461** students receive a 2% bonus for typesetting assignments using $\text{\LaTeX}$.

1. (15 pts) A supersonic flow leaves a two-dimensional nozzle in parallel, horizontal flow (region A) with a Mach number of 2.6 and static pressure (in region A) of 50 kPa. The pressure of the atmosphere into which the jet discharges is 101 kPa. Find the pressures in regions B and C of Figure 1.

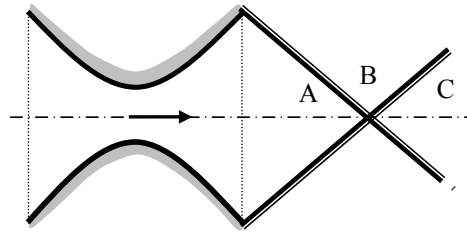


Figure 1: Two-dimensional nozzle with supersonic flow

2. (15 pts) A reservoir containing air ($\gamma = 1.4$) at 2 MPa is connected to ambient air at 101 kPa through a converging-diverging nozzle designed to produce flow at Mach 2.0, with axial flow at the nozzle exit plane (Figure 2). Under these conditions, the nozzle is underexpanded, with a Prandtl–Meyer expansion fan at the exit. Find the flow direction after the initial expansion fan. How does this turning angle affect the net axial thrust forces exerted by the fluid on the nozzle?

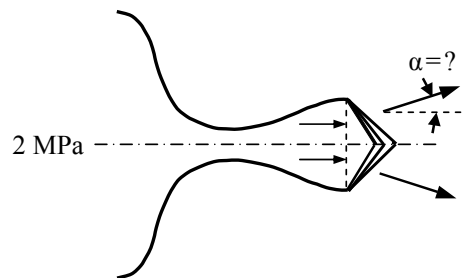


Figure 2: Underexpanded nozzle

3. (40 pts) The symmetrical diamond-shaped airfoil shown in Figure 3 is operating at a 3° angle of attack. The flight speed is $M = 1.8$ and the air pressure equals 0.6 bar.

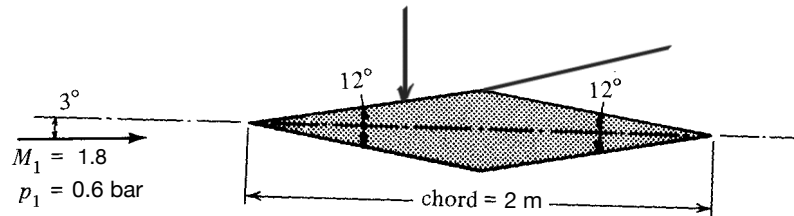


Figure 3: Symmetrical diamond-shaped airfoil

- Compute the pressure on each surface.
 - Calculate the lift and drag forces.
 - Repeat the problem with a 10° angle of attack.
4. (30 pts) **ME 561 ONLY:** A biconvex airfoil (see Figure 4(a)) is constructed of circular arcs. We shall approximate the curve on the upper surface by 10 straight-line segments, as shown in Figure 4(b).

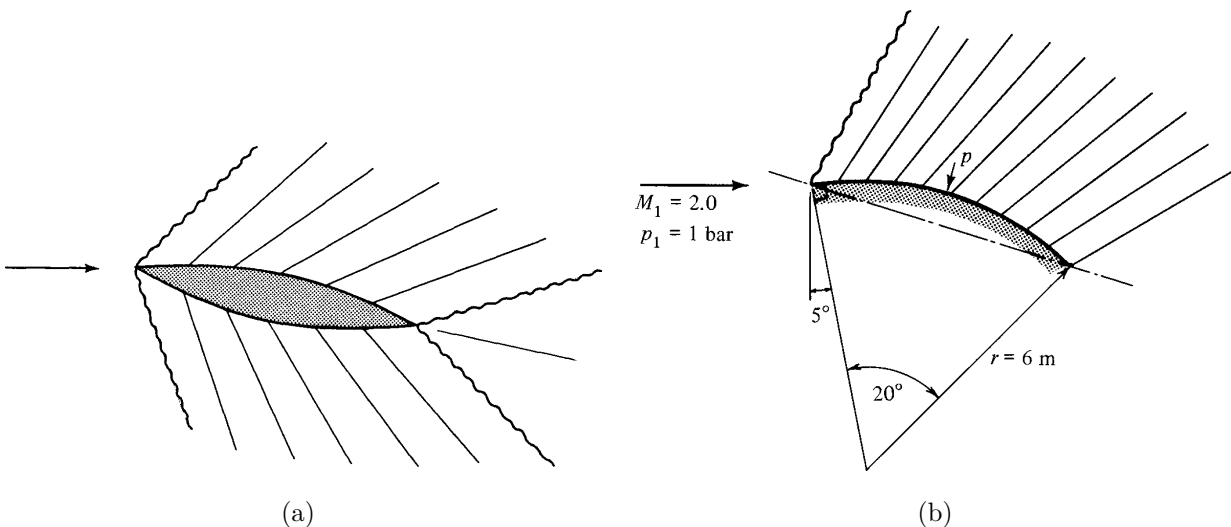


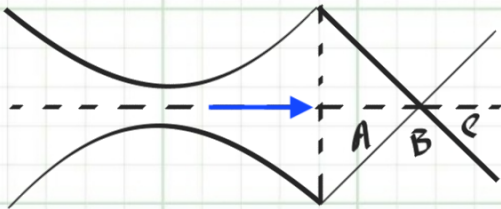
Figure 4: Biconvex airfoil

- Determine the pressure immediately after the oblique shock at the leading edge.
- Determine the Mach number and pressure on each segment.
- Compute the contribution to the lift and drag from each segment.

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Problem 1

$$P_{atm} = 101 \text{ kPa}$$



At A $M=2.6$ with a static pressure of 50 kPa

The jet discharges into atmosphere

Determine

- The Pressures in B + C

$$P_{in B} = P_{atm}$$

$$\frac{P_B}{P_A} = \frac{101 \text{ kPa}}{50 \text{ kPa}} = 2.02$$

$$\Rightarrow \begin{array}{c|c} M_{A1} & M_{B1} \\ \hline 1.37 & 0.7527 \end{array}$$

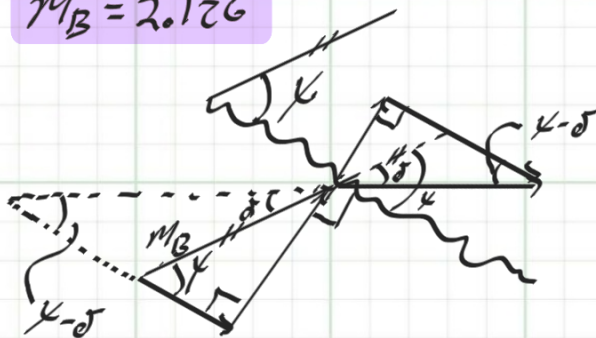
$$\delta = \tan^{-1} \left(\frac{2}{\tan(\theta)} \left(\frac{M_A^2 \sin^2(\theta) - 1}{M_A^2 (\gamma + \cos(2\theta)) + 2} \right) \right) \bigg|_{\substack{M_A = 2.6 \\ \gamma = 1.4 \\ \theta = \sin^{-1} \left(\frac{M_{A1}}{M_A} \right) \bigg| M_{A1} = 1.37}}$$

$$\theta = 31.798^\circ$$

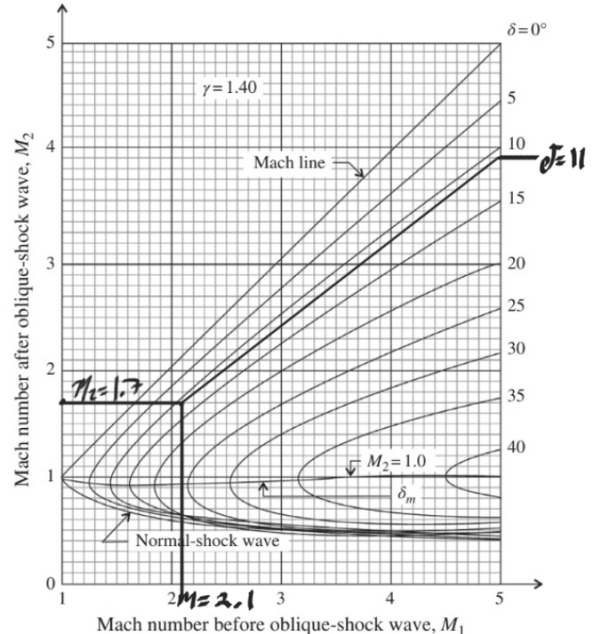
$$\delta = 11.06^\circ$$

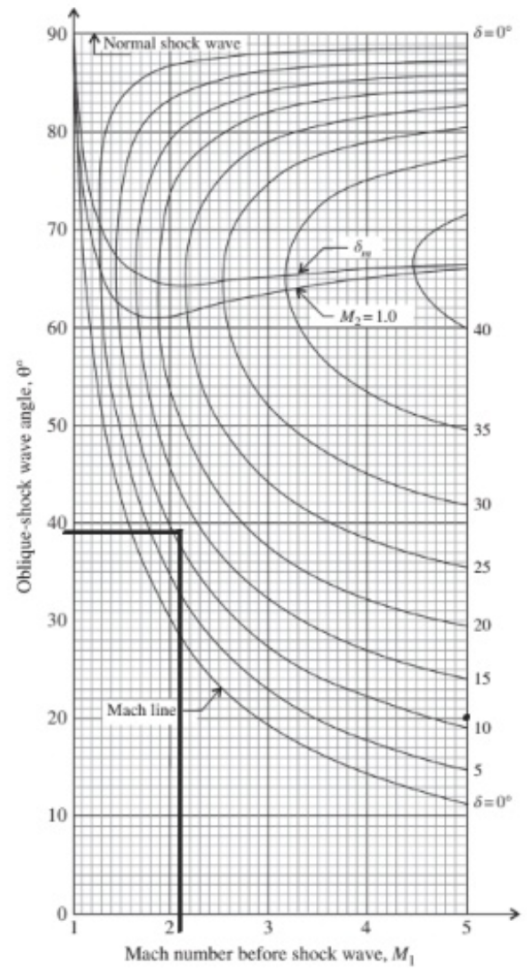
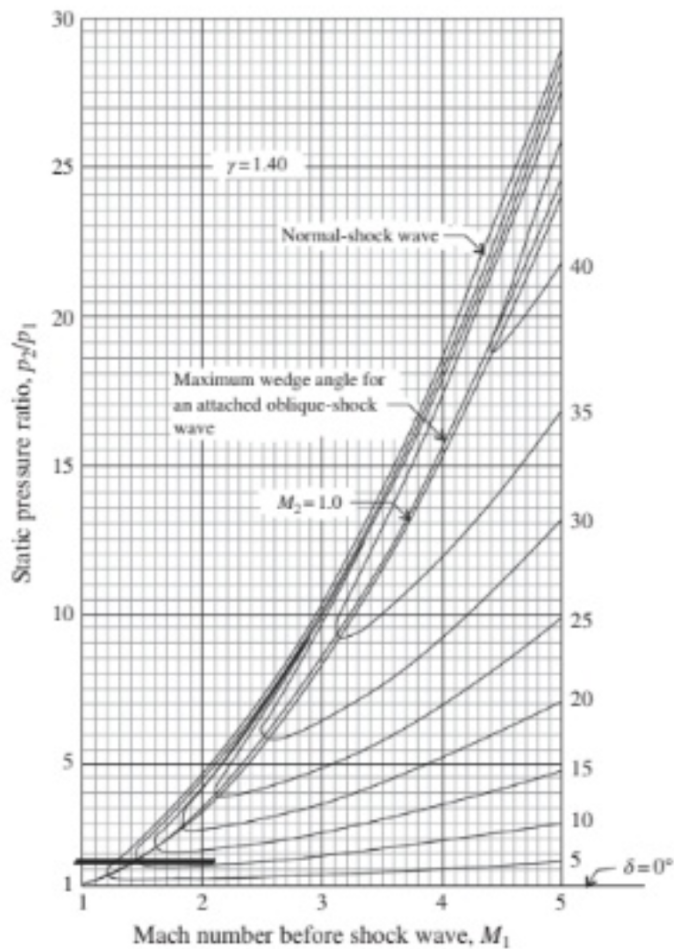
$$\Rightarrow M_B = \frac{M_{B1}}{\sin(\theta - \delta)} = \frac{0.7527}{\sin(20.74^\circ)} = 2.126$$

$$M_B = 2.126$$



Since δ doesn't change $\therefore M_C \neq M_A$
 $\therefore$ using the oblique shock charts
 $M_C \approx 1.7 \therefore$





$$P_2/P_1 \approx 1.75 \mid M_1 \approx 2.1$$

$$\delta \approx 11^\circ$$

$$\Rightarrow P_2 = \frac{P_2}{P_1} P_1$$

$$P_2 \approx (1.75) (101 \text{ kPa})$$

$$P_2 \approx 176.75 \text{ kPa}$$

$$P_1 = 101 \text{ kPa}$$

$$\theta \approx 39^\circ \mid M \approx 2.1$$

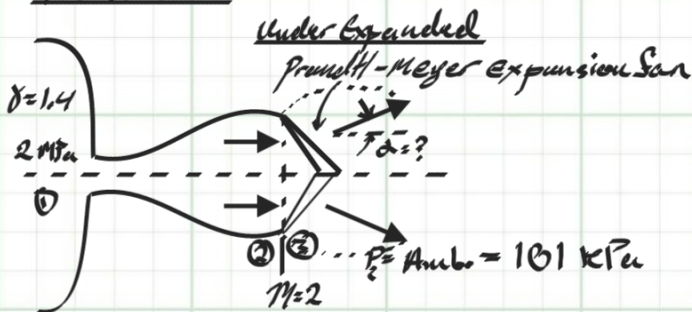
$$\delta \approx 11^\circ$$

$$\text{or from } \theta = 39^\circ$$

$$M_{e1} = M_e \sin(39^\circ)$$

$$M_{e1} = 1.7 \sin(39^\circ) = 1.0698$$

Problem 2

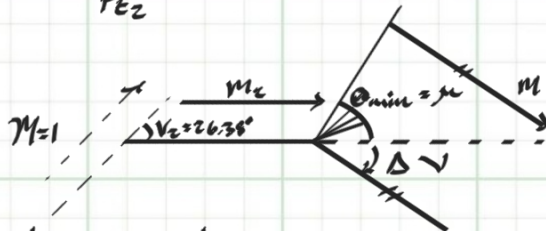


Determine

- flow direction after initial expansion.
- How does this turning angle affect the net axial thrust forces exerted by the fluid on the nozzle?

@ $M = 2$ $\mu = 36^\circ = \theta_{min}$ $\nu = 26.38^\circ$

$$\frac{P_2}{P_{t2}} = (0.1278) \Rightarrow P_2 = (0.1278)(2 \text{ MPa}) = 255.6 \text{ kPa}$$



for $\frac{P_2}{P_{t2}} = \frac{101 \text{ kPa}}{2000 \text{ kPa}} = 0.0505$

$\Delta \nu \equiv$ turning Angle

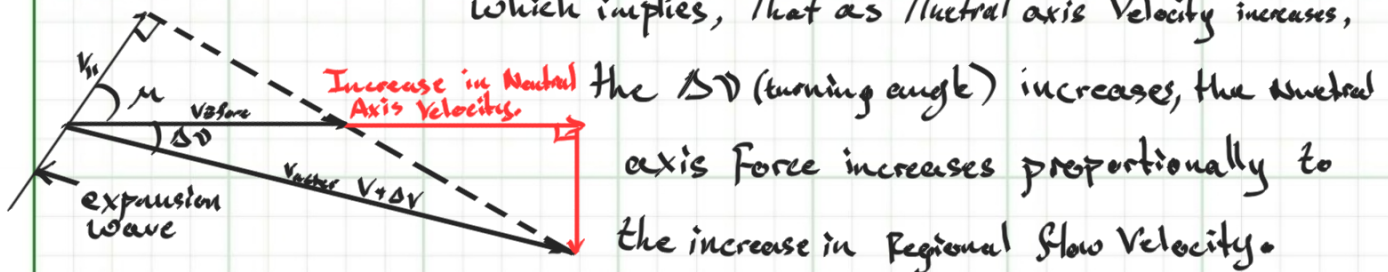
$$\frac{P_3}{P_{t3}} = \frac{P_2}{P_2} \frac{P_2}{P_{t2}} \frac{P_{t2}}{P_{t3}} = \left(\frac{101 \text{ kPa}}{255.6 \text{ kPa}} \right) (0.1278) (1) = 0.0505 \quad \checkmark$$

@ $\frac{P_3}{P_{t3}} = 0.0505$ $M_3 \approx 2.6 \Rightarrow \nu_3 = 41.414^\circ$

$$\therefore \Delta \nu = \nu_3 - \nu_2 = 41.414^\circ - 26.38^\circ = 15.03^\circ$$

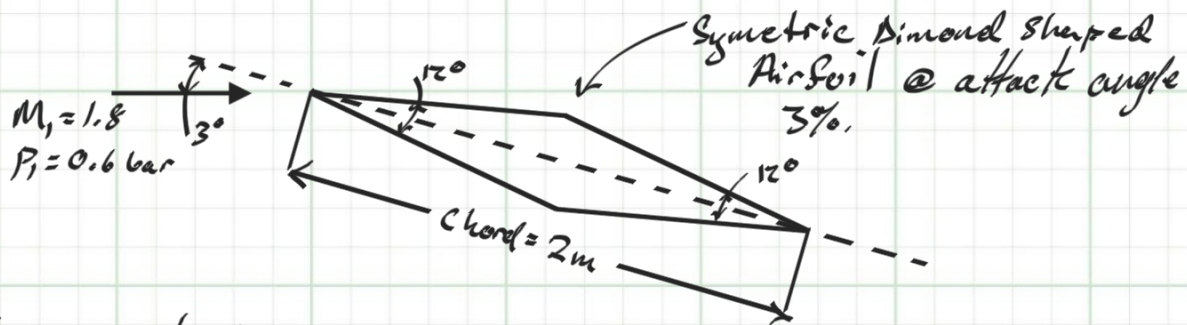
- The flow after the initial expansion is deflected away from the from Neutral flow axis by 15.03°
- As the flow is over Expanded the flow Velocity after the expansion causes a Neutral axis flow Velocity increase

which implies, That as Neutral axis Velocity increases,

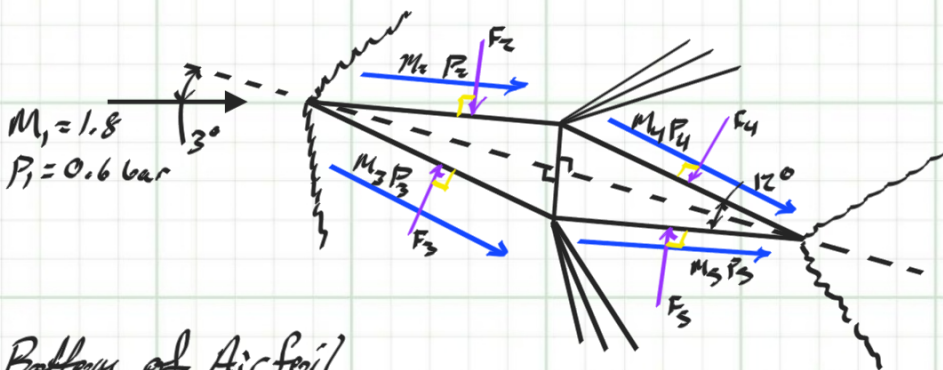


the $\Delta \nu$ (turning angle) increases, the Neutral axis force increases proportionally to the increase in Regional flow Velocity.

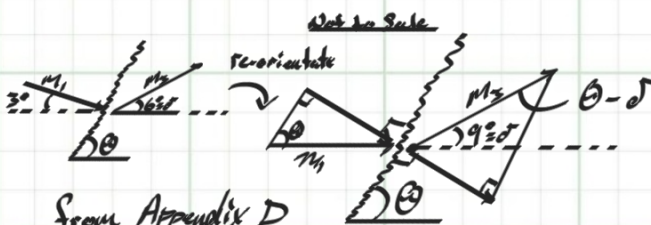
Problem 3



- Compute the pressure on each surface
- Calculate the lift and drag forces
- Repeat A-B w/ 10° angle of attack.



Bottom of Airfoil

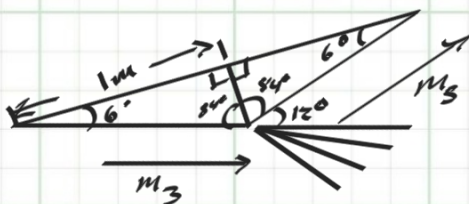


From Appendix D
 $\theta \approx 44^\circ$

$$M_{1n} = M_1 \sin(44^\circ) = 1.25 \Rightarrow M_{3n} = 0.813 \quad \leftarrow \text{N.S. Table}$$

$$\Rightarrow M_3 = M_{3n} \sin(35^\circ) \approx 1.42$$

$$\Rightarrow \frac{P_3}{P_1} = 1.66 \Rightarrow P_3 = 0.996\text{ bar}$$



From Iso/P Prandtl Flow Table

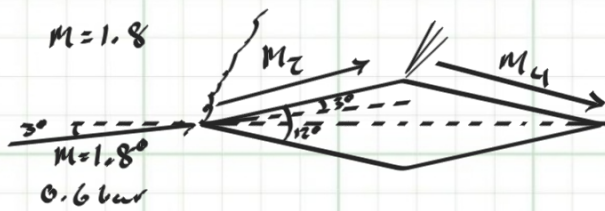
$$@ M_3 = 1.42 \quad V_3 = 9.565$$

$$\frac{P_3}{P_{t3}} = 0.305$$

Since we are turning through the Expansion Wave $V_5 = V_3 + 12^\circ$

$$V_5 \approx 11.565^\circ \Rightarrow M_5 \approx 1.83 \quad \Rightarrow \frac{P_5}{P_{t5}} = 0.16622$$

$$P_5 = \frac{P_3}{P_{t3}} \frac{P_{t3}}{P_5} P_3 = (0.16622 \times 0.305) (0.996\text{ bar}) = 0.543\text{ bar}$$



w/ δ of 3° from Appendix D

$$\theta \approx 37^\circ$$

$$M_{1u} = M_1 \sin(37^\circ) = 1.083$$

$$M_{1u} = 1.083 \Rightarrow M_{2u} = 0.92771$$

$$P_2/P_1 = 1.19415$$

$$M_2 = M_{2u} \sin(34^\circ) = 1.66$$

$$\Rightarrow P_2 = 0.7165 \text{ bar}$$

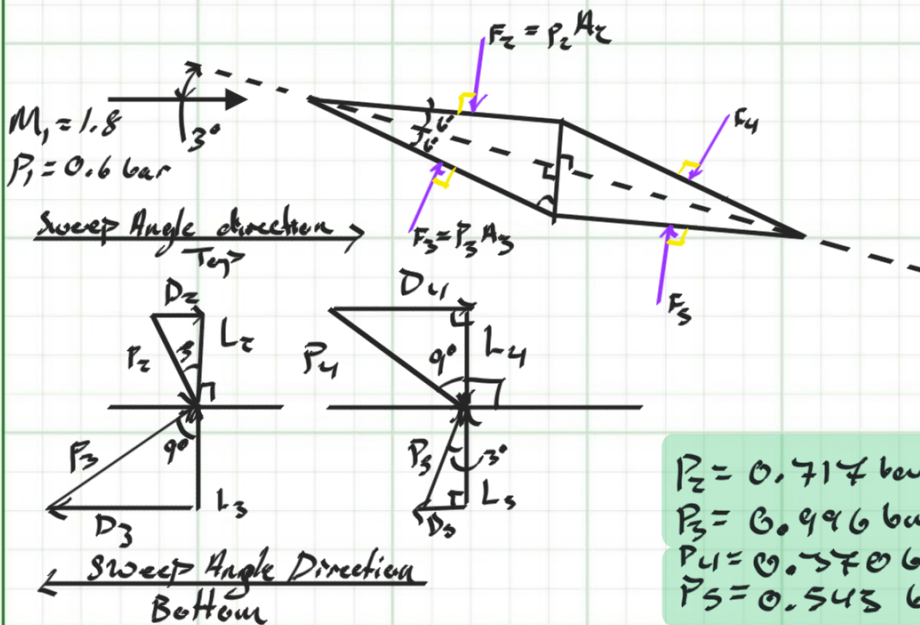
$$@ M_2 = 1.66, \gamma_2 = 16.63284, P/P_e = 0.21515$$

$$\therefore \Delta V = 12^\circ \Rightarrow V_4 = 28.63284 \Rightarrow M_4 = 2.09$$

$$@ M_4 = 2.09, P/P_e = 0.11107$$

$$P_4 = \frac{P_4}{P_e} \frac{P_e}{P_2} P_2 = (0.11107)(0.21515)(0.7165 \text{ bar})$$

$$P_4 = 0.370 \text{ bar}$$



$$P_2 = 0.717 \text{ bar}$$

$$P_3 = 0.996 \text{ bar}$$

$$P_4 = 0.370 \text{ bar}$$

$$P_5 = 0.543 \text{ bar}$$

a)

top P

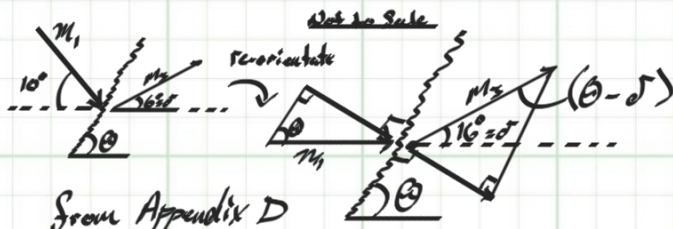
Bottom P

$$\frac{F}{A} = P \Rightarrow L = (P_3 - P_4) \cos(9^\circ) + (P_2 - P_5) \cos(3^\circ) \text{ m}$$

$$\frac{L}{\text{unit span}} = 0.474 \text{ bar.m} = 47.4 \frac{\text{kN}}{\text{m}} \text{ lift per unit-span. } b)$$

$$\frac{D}{m} = (P_4 - P_3)(1\text{m}) \sin(9^\circ) + (P_2 - P_5)(1\text{m}) \sin(3^\circ)$$

$$\frac{\text{Drag}}{\text{unit span}} = -0.0888 \text{ bar.m} = -8.88 \frac{\text{kN}}{\text{m}} \text{ drag per unit-span. } c)$$



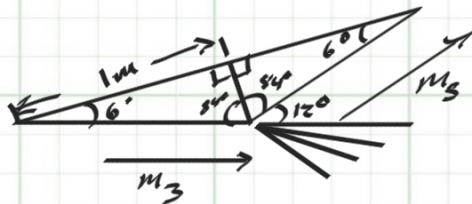
From Appendix D

@ 54° for $M=1.8$ & $\theta=16^\circ$

$$M_{2n} = M_1 \sin(54^\circ) \approx 1.46 \Rightarrow M_{3n} = 0.71374$$

$$\Rightarrow M_3 = M_{3n} \sin(38^\circ) \approx 1.163$$

$$\Rightarrow \frac{P_3}{P_1} = 2.32 \Rightarrow P_3 = 1.392 \text{ bar}$$



From Iso/Praandtl. Flow Table

$$@ M_3 = 1.163 \quad \gamma_3 = 2.60735$$

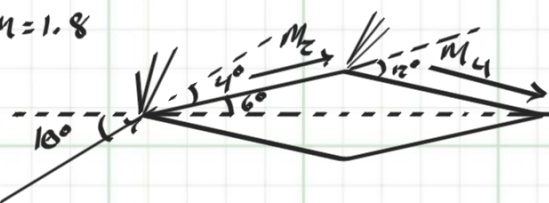
$$\frac{P_3}{P_{t3}} = 0.4343$$

Since we are turning through the Expansion Wave $\gamma_5 = \gamma_3 + 12^\circ$

$$\gamma_5 = 14.607 \Rightarrow M_5 \approx 1.59 \Rightarrow \frac{P_5}{P_{t5}} = 0.23878$$

$$P_5 = \frac{P_3}{P_{t5}} \frac{P_{t3}}{P_3} P_3 = (0.23878)(0.4343)(1.392) = 0.765 \text{ bar}$$

$M=1.8$



$$@ M=1.8 \quad \gamma = 20.725$$

$$\Rightarrow \gamma_2 = \gamma + \Delta\gamma \mid \Delta\gamma = 4^\circ \Rightarrow \gamma_2 = 24.725^\circ$$

$$M_2 = 1.44 \quad \frac{P_2}{P_{t2}} = 0.14028$$

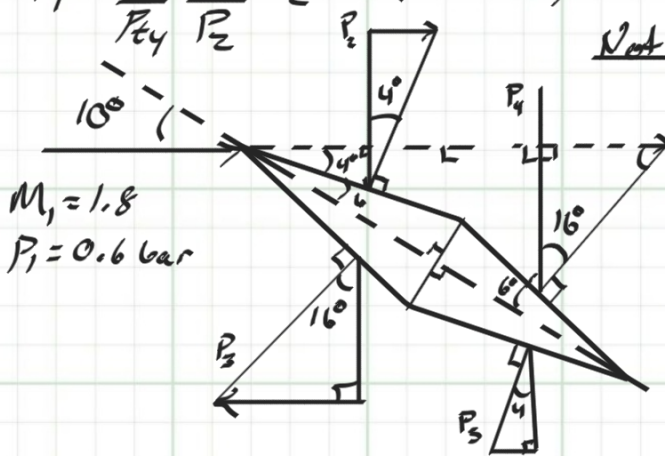
$$\gamma_4 = \gamma_2 + \Delta\gamma \Rightarrow \gamma_4 = 24.725^\circ + 12^\circ = 36.73^\circ$$

$$@ M=1.8^\circ \quad \frac{P_1}{P_{t1}} = 0.17404$$

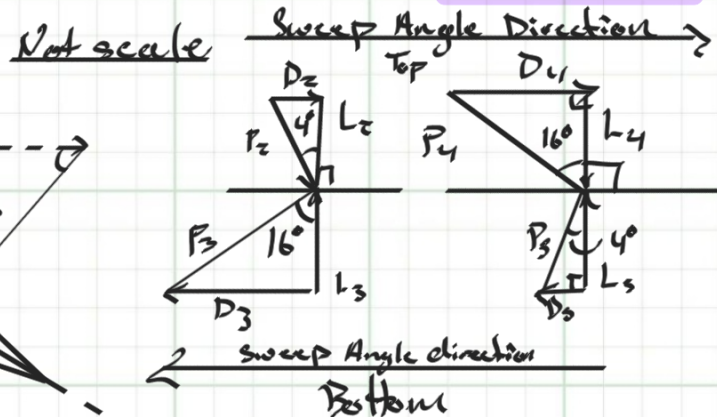
$$@ \gamma = 36.73^\circ \Rightarrow M_4 = 2.4, \frac{P_4}{P_{t4}} = 0.06840$$

$$P_2 = \frac{P_3}{P_{t2}} \frac{P_{t1}}{P_1} P_1 = (0.14028)(0.17404)(0.6 \text{ bar}) = 0.4836 \text{ bar}$$

$$P_4 = \frac{P_4}{P_{t4}} \frac{P_{t2}}{P_2} P_2 = (0.06840)(0.14028)(0.4836) \text{ bar} = 0.2358 \text{ bar}$$



$M_1 = 1.8$
 $P_1 = 0.6 \text{ bar}$



$$\begin{aligned}
 P_2 &= 0.484 \text{ bar} \\
 P_3 &= 1.392 \text{ bar} \\
 P_4 &= 0.236 \text{ bar} \\
 P_5 &= 0.765 \text{ bar}
 \end{aligned}$$

$$\frac{F}{A} = P \Rightarrow L = (P_3 - P_4) \cos(16^\circ) + (P_5 - P_2) \cos(4^\circ) \text{ m}$$

$$\frac{L}{\text{unit-span}} = 1.436 \text{ bar} = 143.6 \frac{\text{kN}}{\text{m}} \text{ lift per unit-span.}$$

$$\frac{D}{\text{unit-span}} = ((P_4 - P_3) \sin(16^\circ) + (P_2 - P_5) \sin(4^\circ)) \text{ m}$$

$$\frac{\text{Drag}}{\text{unit-span}} = 0.338 \text{ bar m} = -33.8 \text{ kN/m} \text{ drag per unit-span.}$$